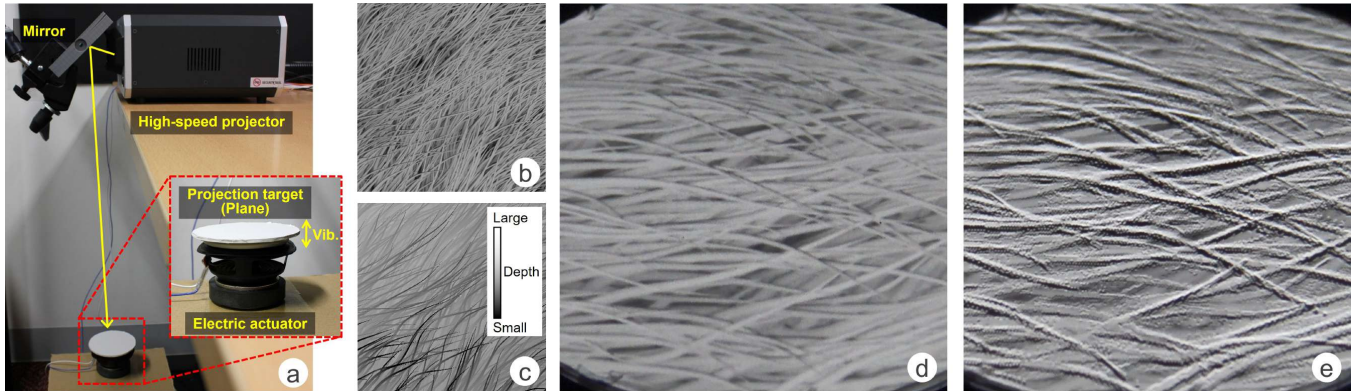


# 3D Texture Representation in Projection Mapping onto a Surface with Micro-Vibration

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**Figure 1:** (a) The implemented system. (b) The original image to be projected, which represents a fur. (c) The depth image, which represents the depth value of each pixel in the original image. (d) The result of simply projecting the original image onto a static plane. (e) The result of projecting the original image by the proposed method. Compared to the simple projection, the fur texture is represented three-dimensionally on the vibrating flat surface.

## Abstract

In projection mapping, it is not possible to adequately represent fine, three-dimensional surface textures such as fur. To solve this problem, we propose a method to add minute high-speed vibrations in the depth direction to the projection target and to project images at high speed synchronized with the depth position of the projection target at each instant. In this paper, we report on the proposed method and its projection results.

## CCS Concepts

• Computing methodologies → Mixed / augmented reality.

## Keywords

Projection mapping, 3D representation, texture

## ACM Reference Format:

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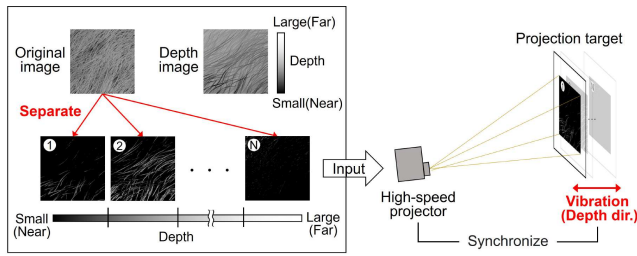
<https://doi.org/10.1145/3681756.3697944>

## 1 Introduction

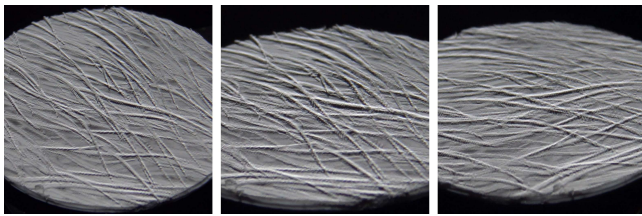
In projection mapping, the appearance of the projection target, such as its color and texture, can be changed [Raskar et al. 2001]. This technique is useful in that it can reproduce various appearances on a real object without physically preparing the desired surface texture. However, images projected from a projector are perceived along the surface shape of the projection target, and thus cannot adequately represent three-dimensional textures such as fur.

To address this problem, there is research that projects images that take into account the parallax of both human eyes to give the perception of a three-dimensional shape on the surface of the projection target [Okutani et al. 2018]. However, this research requires the wearing of special glasses, which undermines the advantage of projection mapping that images can be viewed without wearing a device. In addition, it does not deal with the three-dimensional perception of microshapes, such as fur.

Therefore, the purpose of this research is to make a texture with a fine shape like fur be perceived as three-dimensional on the surface of the projection target. For this purpose, we propose a method of adding minute high-speed vibrations in the depth direction to a projection target and projecting images at high speed synchronized with the depth position of the projection target at each instant. The method takes advantage of the visual property that light flashing faster than the critical fusion frequency of the human eye is perceived as averaged [Nelson and Bartley 1964].



**Figure 2: Process of the proposed method.** The left frame shows how the input images are generated. The original image is separated into multiple layers based on its depth image. These generated images are input and projected at high speed in synchronization with the high-speed vibration of the projection target in the depth direction.



**Figure 3: The projection results of the proposed method taken from different viewpoints.** It can be seen that the image appears to be three-dimensional even if the observer's position changes.

## 2 Method

The process of the proposed method is shown in Figure 2. First, to generate the input image, the original image to be represented and its depth image are prepared. Next, the original image is separated into multiple layers based on the depth image. In other words, the original image is separated at regular intervals in the depth direction.

The generated separated images are then projected in synchronization with the depth position of the projection target at each instant, which vibrates at high speed in the depth direction. When the projection target comes closest, only the foremost separated image with the smallest depth value is projected, and as the projection target moves away, only the more distant separated images with larger depth values are projected one by one. This means that the smaller the depth value of the separated image is, the more the image is formed in the front, and the larger the depth value of the separated image is, the more the image is formed in the back. By doing this periodically at high speeds above the critical fusion frequency, the human eye averages the images and perceives them as a single image. The observed image is perceived as if it were three-dimensional, because at each instant, only the separated images with depth values corresponding to the depth position of the projection target are projected.

## 3 Result

The implemented system based on the proposed method is shown in Figure 1(a). A high-speed projector (manufactured by Tokyo Electron Device Limited, TB-UK-DYNAFLASH) with a maximum frame rate of 2,841 fps was used for image projection. This projector is capable of projecting 8-bit grayscale images at XGA resolution. The image to be projected is switched by inputting an external trigger signal to the projector synchronized with the vibration of the projection target. A voice coil-type electric actuator was used to add vibration to the projection target. The projection target is fixed to the voice coil, and the target is vibrated uniaxially by controlling the coil current. When the voice coil is used to vibrate the projection target at 60 Hz, which is the critical fusion frequency of the human eye, the vibration width is about 2.0 mm.

The result of the projection by the proposed method with this implemented system is shown in Figure 1(e). Figure 1(b) is the original image of the fur texture to be represented, and its depth image in Figure 1(c) was used to separate the original image into 16 layers before projection. Figure 1(d) shows the result of simply projecting the original image onto a stationary plane. Compared to the result of simple projection, the result of the proposed method confirms that the fur texture in the original image is perceived as three-dimensional. Also, Figure 3 shows the projection results of the proposed method viewed from various angles. As shown in this figure, the projection result appears three-dimensional even if the observer's eye position changes.

## 4 Conclusion

In this paper, we focused on the fact that the image represented in projection mapping depends on the surface shape of the projection target, and aimed to represent the projection result as if it were three-dimensional. To achieve this, we proposed a method to vibrate the projection target slightly and at high speed in the depth direction and to project images at high speed synchronized with the depth position of the projection target at each instant. As a result, it was shown that this method can give a three-dimensional perception to the projection result. Furthermore, it was also shown that the projection result was perceived as three-dimensional even if the observer's eye position changed.

However, the proposed method causes jagged artifacts in the projected image and visual discomfort such as flickering to the observer. We believe that these can be reduced by removing high-frequency components and improving the projection speed, which we will address in the future.

## Acknowledgments

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